Persona-Aware, Knowledge-Bounded NPC Dialogue Through Model-Agnostic LLM Orchestration

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***Abstract*—Conversational non-player characters must maintain character roles while distinguishing available game knowledge from unsupported player claims. This paper presents Anima, a policy-controlled dialogue architecture integrating persona appraisal, deterministic engagement, source-bound retrieval, compatible model invocation, and bounded response acceptance. External documents are indexed using CPU MiniLM embeddings and Chroma, while a common provider contract separates orchestration from inference backends. Evaluation combines controlled retrieval, a matched real-model dialogue comparison, an index update, and a native Skyrim integration test. Across 24 authored queries over 12 documents, Precision@1, Recall@3, and mean reciprocal rank were 1.00; relevance labels lack independent adjudication. Seven completed matched pairs contained predefined answer targets in seven retrieval-enabled responses and one retrieval-disabled response. This literal diagnostic does not measure semantic faithfulness. Thirty warm retrieval trials averaged 19.96 ms, and synchronization made a revised fact retrievable without a generator-training operation. In Skyrim 1.7.104, the native connector acquired NPC context and acknowledged completion of an authored text response. These preliminary observations demonstrate implemented knowledge-access and delivery mechanisms while leaving independent persona evaluation, semantic containment, and end-to-end gameplay performance unresolved.**

***Index Terms*—non-player characters, persona conditioning, retrieval-augmented generation, model abstraction, knowledge boundaries, dialogue evaluation**

I. Introduction

Non-player character (NPC) dialogue must respond to a player's request without abandoning the character's identity, obligations, or knowledge. Authored dialogue gives predictable control, but broader coverage requires additional content and branching logic. Language models supply flexible realization; fluency alone, however, does not establish that an NPC knows a fact or should engage. Game-dialogue research already motivates conditioning on character, history, and state [1].

Game knowledge changes independently of model behavior: locations and character facts may be revised while an inference backend remains fixed. Character training [2] and parameter-efficient adaptation [3] remain legitimate strategies. External retrieval [4] offers a complementary maintenance mechanism by supplying updated domain facts during inference rather than requiring an application training operation.

Anima places a dedicated orchestration layer between interaction interfaces and model inference. The layer appraises persona and trusted state, chooses authored/end/generated routes, retrieves source-bound evidence when needed, and validates model output under deadlines. Its contribution is the integration of these boundaries, not the invention of RAG, persona prompting, or provider abstraction. The term knowledge-bounded describes the architecture's intended access and acceptance controls; it does not assert guaranteed semantic faithfulness.

Our contributions are the implemented policy-controlled composition, controlled retrieval and matched dialogue evidence, and correlated native Skyrim delivery. Small authored datasets, quota-limited generation and missing independent ratings constrain claims about quality and generalization.

II. Related Work

RAG combines parametric generation with nonparametric retrieved memory [4]. Anima supplies passages to an external inference model rather than training the RAG components. Dense passage retrieval establishes representation-based candidate selection [5]; our implementation uses an existing MiniLM encoder and cosine search, not a new retrieval objective.

Character-LLM adapts characters through profiles and experiences [2]. Anima instead keeps authored personas and current state outside generator parameters. This does not establish superior character quality or training-cost savings; LoRA [3] is a relevant efficient adaptation alternative.

Generative Agents already orchestrates memory, retrieval, reflection, and planning [6]. Consequently, the defensible contribution here is a particular policy-controlled integration with explicit trust, evidence, and timing boundaries, rather than a first unified agent framework. Game-state-aware dialogue is also established [1]. Anima focuses on reusable runtime composition, without reproducing autonomous town simulation or a trained gameplay model.

Profile-conditioned dialogue is established by Persona-Chat [7]; an authored profile alone is neither a new method nor evidence of sustained persona consistency. Anima adds runtime source authorization, deterministic route selection and game delivery around that conditioning.

Evaluation must separate retrieval relevance, faithful evidence use, and generation quality, as emphasized by RAGAs [8]. Long-context availability does not guarantee reliable context utilization [9]. Our literal answer-target diagnostic therefore cannot substitute for independent claim-level support or persona evaluation.

III. Problem Formulation and Research Contributions

For utterance u, trusted snapshot s, observations o, persona p, and external documents D, appraisal constructs $c = A(u, s, o, p)$ and policy selects $a = \pi(c)$. A generated route retrieves $E = R(q(c), D, b)$ under trusted source binding b, constructs prompt $P = B(c, a, E)$, and obtains candidate $y = G_\theta(P)$. Acceptance $V(y, c, a, E)$ controls whether the response leaves the runner. Here q resolves the retrieval question, b is the authorized source binding, B packs the prompt, and G_θ is the selected generator. Authored/end routes bypass retrieval and inference.

We ask: **RQ1**, how well does retrieval rank predeclared relevant sources in a controlled corpus? **RQ2**, how does enabling the implemented RAG path change literal answer-target presence in matched model outputs? **RQ3**, which persona and knowledge-access controls are implemented, and what semantic properties remain unevaluated? **RQ4**, where can compatible inference backends be substituted without changing orchestration? **RQ5**, can an edited fact become retrievable after synchronization without an application generator-training step? RQ1, RQ2 and the retrieval-level portion of RQ5 have measured results; RQ3 and RQ4 are implementation assessments, not semantic-quality or multi-provider performance experiments.

The implementation contribution comprises deterministic engagement before model work, separation of trusted observations and untrusted references, source-bound retrieval with prompt-level evidence accounting, bounded candidate acceptance through a provider-neutral interface, and revision-correlated native game delivery. Model agnosticism is restricted to that interface and compatible transports. Incompatible APIs require an adapter, and model capabilities remain consequential.

IV. Proposed System Architecture

Fig. 1 presents the implemented component boundaries. The central `PreviewRunner` composes policy, context, optional retrieval, provider invocation, validation, and one permitted repair. The local lab and Skyrim connector invoke the same core rather than separate dialogue engines.

Appraisal verifies scope/revision and filters observations by perceiver, subject, confidence, and freshness. Conflicting equal-rank observations remain unknown. Persona and relationships influence tone and willingness, while familiarity does not grant knowledge. Deterministic policy can select authored greetings/refusals or end interaction before calling a model.

Retrieval is restricted by configured game/world/NPC file bindings; approximate similarity does not itself authorize a source. Prompt packing rechecks evidence and preserves whole passages. The provider contract contains messages, text, model identity, finish reason, and usage rather than vendor-specific orchestration policy. Candidate output cannot directly mutate quests, inventory, or relationships.

The lab's SQLite slice stores revision-checked relationships and bounded display-acknowledged history. Display acknowledgement is not proof of hearing. The Skyrim bridge has correlated native text/audio commands and best-effort receipts; a durable production outbox and complete save-history reconstruction are outside this slice.

V. System Methodology

A. Offline Knowledge and Online Retrieval

Loaders extract text from supported plain-text, structured, PDF, DOCX, and HTML files; OCR is not implemented. Encoder token offsets form default 220-token chunks with 32-token overlap. The all-MiniLM-L6-v2 sentence encoder [10], [11] runs through ONNX Runtime [12] and produces 384-dimensional CPU vectors. Chroma [13] stores cosine-search vectors with source/revision metadata. File hashes include content and pipeline parameters; a manifest publishes a replacement only after that file's batches succeed. Index synchronization is explicit, outside dialogue turns.

Online retrieval embeds the resolved question and searches committed revisions of the authorized source. At most three evidence passages enter the dialogue boundary. No learned reranker is implemented. Truncated, duplicate, empty, wrong-source, or mismatched evidence is removed or rejected. Optional distance filtering is not calibrated confidence. The manifest is enumerated to form revision filters, so large-corpus authorization scaling remains unmeasured.

B. Persona, Prompt, and Provider Boundary

Fig. 2 traces online route selection, retrieval, candidate acceptance and delivery. System content expresses authored persona, plan, response limits, and knowledge instructions. Player text, retrieved passages, history, and visual descriptions remain data. This separation is a prompt boundary, not a proven injection defense. Whole evidence passages precede recent whole exchanges during packing; short citation aliases resolve only to packed identifiers. A conservative UTF-8 byte estimate is used instead of exact backend tokenization.

Fig. 3 distinguishes the common interface, compatible HTTP transport, profile configuration, normalized reply, and offline double. Gemini and OpenRouter profiles use one chat-completions transport; they are not unrelated native SDK adapters. The scripted provider is not an LLM and is excluded from scientific model comparisons.

Validation checks current state, finish reason, strict JSON/schema, modes, length, cues, and evidence-ID membership. Membership does not imply entailment, and the validator does not require support for every factual claim. One repair may occur within the same cutoff. Failures can select a validated authored fallback; stale/cancelled work produces no new speech. Local cancellation cannot guarantee cancellation of remote billing.

VI. Implementation and Experimental Setup

Controlled retrieval and model experiments use source baseline ff9fd7e. The separate native delivery test uses Anima-Connector 0.2.0 on Skyrim 1.7.104.0; it does not alter those

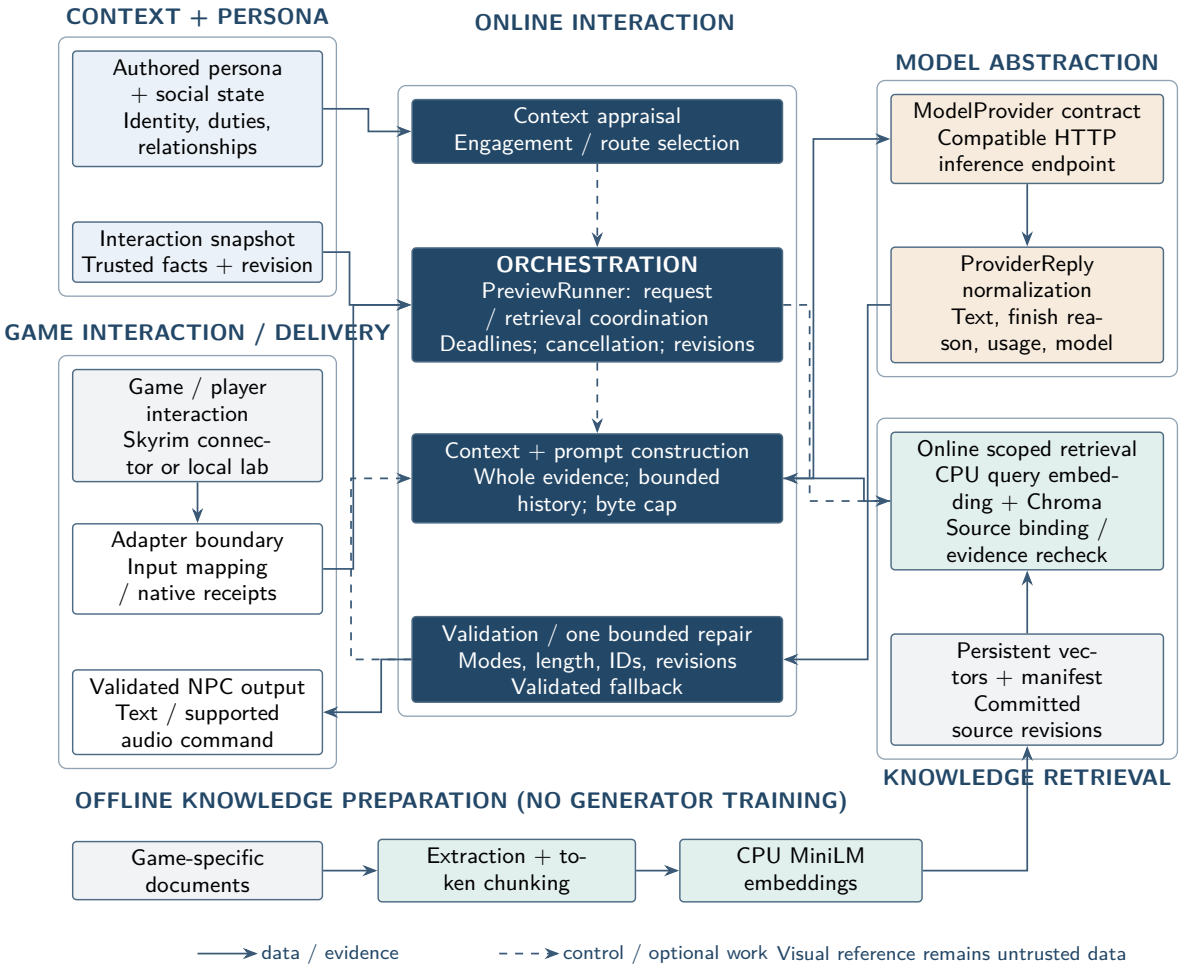


Fig. 1. Policy-controlled architecture linking game interaction, knowledge access and response acceptance. Grouped boundaries separate interaction, persona/context, central orchestration, retrieval and provider normalization. Solid arrows carry data/evidence; dashed arrows denote control or optional work. Offline preparation is separate from inference. Native delivery is evaluated separately from retrieval and inference quality.

measurements. Experiments use the project Python 3.11.9 environment, Chroma 1.5.9, ONNX Runtime 1.23.2 and HTTPX 0.28.1 on Windows. Hardware reports an Intel Core i7-13700HX, 16 cores/24 logical processors and approximately 15.7 GiB physical memory. These are observed conditions, not deployment requirements.

A. *Datasets and Metrics*

We authored Cedarvale, a controlled fictional village corpus of 12 short topic documents, yielding 12 chunks, and 24 paraphrased queries. Each query has one predeclared relevant source. Labels were fixed before retrieval and are independent of ranking output, but were assigned by the experiment author, not independent human adjudicators. This is not a Skyrim corpus or a natural-query benchmark. We report Precision@1, Precision@3, Recall@3, and reciprocal rank; missing positions count incorrect. With one relevant source, Precision@3 has ceiling $1/3$.

A generation corpus concatenates those documents into one authorized public file; a second fictional private-code file is excluded by binding. Thus, retrieval and generation use the same authored facts but different chunk layouts. Twelve planned prompts span four authored personas, answerable facts, missing information, false premises and restricted knowledge. Only seven complete answerable pairs and one additional RAG trial finished before rate limiting. Boundary cases were not reached.

Paired trials invoke the actual PreviewRunner and configured gemini-3.5-flash-lite compatible endpoint. Persona, utterance, relationship, model and caps remain matched; RAG mode adds its implemented evidence instructions as well as passages. This is an ablation of the implemented RAG path, not an isolated effect of document text alone. Order alternates by case; one sample per condition is collected without a deterministic sampling seed. Caps are 12,000 input bytes, 400 output tokens, 30-s transport, and a development 60-s response deadline with 59-s model cutoff.

Eight planned answerable prompts have literal answer targets fixed before calls. Target presence is case-insensitive substring matching, not semantic correctness. Independent persona, relevance and claim-support ratings were not collected;

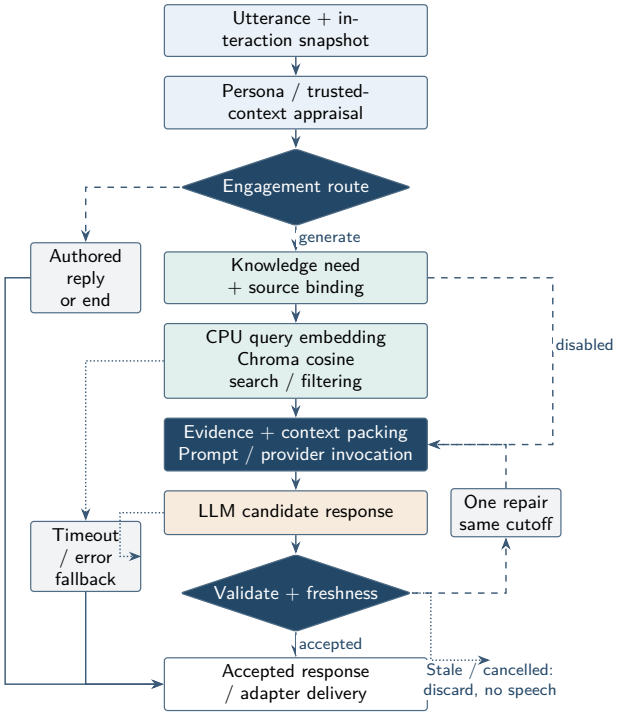


Fig. 2. Implemented online branches, including authored/end bypass, optional retrieval, bounded repair, timeout fallback and stale/cancelled rejection. Repair shares the original cutoff.

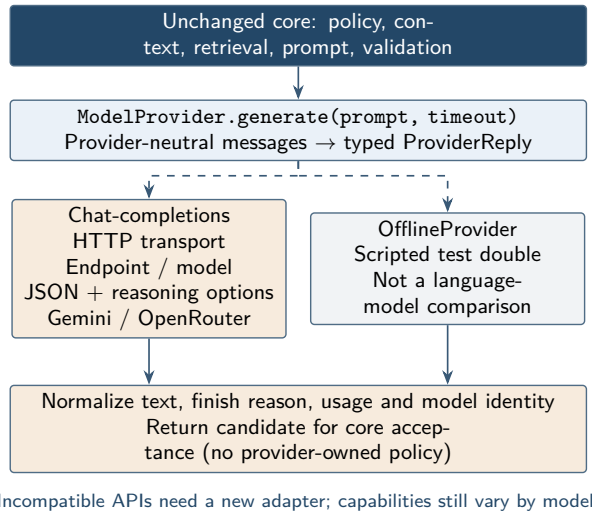


Fig. 3. Provider independence at the implemented contract. Backend substitution preserves core logic while endpoint/model/format settings can change. Unsupported protocols need new adapters; offline doubles are distinct from actual inference.

blank blinded sheets accompany the artifacts.

B. Timing and Update Protocol

Thirty repeated warm retrieval trials cycle through the 24 queries, recording query-embedding, vector-query and total retrieval time with a monotonic clock. One fresh-process query measures first-use loading separately; it is not a cold-start

TABLE I
EVALUATION COVERAGE AND QUANTITATIVE SUMMARY. SEPARATE EXPERIMENTS RETAIN THEIR OWN DENOMINATORS.

Evidence unit	Observed value
Controlled model questions / personas	8 / 4
Completed matched questions / generated outputs	7 / 14
No-RAG / RAG accepted generated outputs	7 / 7
Additional RAG error-fallback outputs	1
Provider requests (including repair attempts)	17
Native greeting strings / named NPCs	2 / 2
Retrieval queries / controlled documents	24 / 12
Retrieval P@1 / R@3 / MRR	1 / 1 / 1
Retrieval P@3 (one relevant source/query)	0.333
Warm retrieval mean ($n = 30$)	19.96 ms
Knowledge synchronization (one update)	216.60 ms

distribution. The model study records response-ready timing, raw candidates, usage and failure paths. Skyrim was running in the background during model trials; network and game contention were not controlled. End-to-end game delivery, image acquisition and frame-time latency distributions were not measured; the native test below evaluates delivery lifecycle only.

The update experiment changes the forge location from "northern well" to "southern mill" in a disposable document copy, synchronizes the index, and repeats source-bound retrieval. Before/after passage IDs and text are preserved. There is no application training operation. A post-update generated answer was planned but not reached after the provider's rate limit; remote parameter hashes are inaccessible.

VII. Results and Discussion

A. Dialogue Evaluation and Test Coverage

The model study reached eight questions across Mira (blacksmith), Iona (healer), Elin (innkeeper) and Daren (guard): seven completed both conditions and one produced an error fallback. Table I separates inputs, routes and evaluation units. Table II reproduces all 14 matched outputs in full against targets fixed before generation; "match" denotes substring presence.

Coverage includes locations, services, item knowledge, identity and guard procedure. Four planned boundary prompts were not reached; relationship-conditioned recall, continuity and prompt injection remain untested by this model sample. Native "Hello" and "HELLO" exercise authored routing separately. Counting those two strings yields ten distinct inputs overall; repeated delivery attempts are not new inputs, and neither fallbacks nor receipts count as generated successes.

B. Retrieval Effectiveness

All 24 queries ranked their declared relevant source first: Precision@1, Recall@3 and mean reciprocal rank were 1.00; Precision@3 was 0.333, its single-relevant-source ceiling. The controlled corpus isolates whether query embedding, indexing and source selection recover predeclared facts. Correct ranking

across these paraphrases supports that mechanism (RQ1), but does not establish robustness to noisy corpora or natural-game questions. Independent label adjudication and larger held-out corpora are needed for generalization.

C. RAG-Enabled Dialogue Comparison

Table III summarizes the seven complete matched pairs. Retrieval-enabled outputs mentioned their predefined target in seven cases, versus one without retrieval. For Mira's forge, retrieval supplied "northern well," whereas the disabled path invented a village-center location. The enabled healer response named silverleaf salve, while the disabled response proposed calendula and honey. These examples show how external evidence changed specific outputs; they do not estimate a general hallucination rate.

Table II exposes both agreement and unsupported elaboration in the fictional Cedarvale study. All 14 outputs used the generated route; D4 without RAG required one repair. Native greetings and the unmatched fallback are reported separately, so successful delivery cannot inflate model-quality results.

The disabled guard response contained "permit," meeting its literal target, while adding an unsupported cargo-manifest procedure. Thus, target matching cannot establish factual correctness, abstention or persona consistency (RQ2/RQ3). The contrast also includes the implemented evidence instructions, rather than isolating passage text alone.

The study recorded 15 responses from 17 provider requests: 14 accepted generated outputs, including one successful repair, and one unpaired RAG fallback after an invalid candidate and rate-limited repair. Daren's unmatched input "Where is your checkpoint?" yielded "I cannot speak to that matter." through the authored error-fallback route. It is excluded from paired comparisons and retained in failure accounting. Across all eight RAG records, mean response-ready time was 1395.6 ms. No tail latency is estimated from seven pairs. Only one real model produced study responses; backend substitution (RQ4) is supported by the common contract, not a controlled multi-model comparison.

D. In-Game Integration and Dialogue Delivery

AnimaConnector 0.2.0 was exercised in modded Skyrim 1.7.104.0. Opening the normal dialogue menu identifies an NPC and starts a correlated session. The connector acquires actor identity, equipment, combat state, location/weather and offered dialogue; snapshot refreshes carry relevant changes into canonical context. Observed factions and quest records do not automatically authorize NPC knowledge.

Player text reaches the loopback bridge and the same policy/context pipeline as the preview. Selected or generated output returns with NPC/session identity, snapshot revision, delivery ID and expiry. The native receiver rechecks these fields and game readiness before staging text and optional audio. Correlated receipts distinguish started, completed, interrupted and failed delivery. This flow separates response acceptance from actual receiver processing.

Table IV records the named NPC demonstrations and a separate interruption trial. Jenassa's actor and offered-dialogue context reached the bridge; "Hello" selected "What do you need?" through the authored route. A Whiterun Guard "HELLO" trial selected the same reply, with zero model calls and correlated started/completed receipts. The receipt file for the earlier interrupted greeting lacks an NPC field, so that attempt is not assigned an identity. Only Anima-correlated records are included despite concurrent third-party mods.

These tests establish native context acquisition and authored-response delivery lifecycle, not live LLM/RAG quality or independent confirmation that a player saw or heard the reply. Completion denotes the receiver lifecycle. Voice and frame-time effects were not evaluated. Inspected connector captures contain a black game region, leaving a publication-quality gameplay figure pending; no lab, third-party-mod or synthetic image substitutes for it.

E. Runtime Performance

Table V reports 30 warm local retrieval observations. Total retrieval includes query embedding, the Chroma call and surrounding application work; it is not the sum of only the two listed sub-stages. Warm retrieval's descriptive nearest-rank P95 was 23.00 ms.

One fresh-process query took 257.95 ms, including 225.85 ms in embedding and 31.41 ms in vector querying. This single observation includes first-use loading with downloaded assets and an existing index; it is not a disk-cold latency distribution. Initial indexing, including first model use, took 769.40 ms; knowledge synchronization took 216.60 ms. These values concern a tiny corpus and one machine, not full gameplay acquisition-to-display latency or production guarantees. Model timing used a development deadline rather than the normal 3-s preview target.

F. Knowledge Update Demonstration

The source edit changed Mira's forge location from "northern well" to "southern mill." Synchronization published a new revision and subsequent source-bound retrieval returned the updated passage. No generator-training operation occurred. This demonstrates that externally maintained facts can change through document/index maintenance while retaining the inference interface (RQ5). It does not establish answer-level propagation: a post-update generated answer was not collected, and remote model parameter hashes were inaccessible.

G. Discussion

RQ1 establishes ranking on fixed controlled labels; RQ2 shows preliminary output changes when evidence and its instructions enter inference. RQ3 is supported at access-control and structural-validation level, while unsupported guard elaboration motivates independent semantic evaluation. RQ4 remains an interface property rather than a multi-model performance result. RQ5 is demonstrated at retrieval level, not generated-answer level. Native receipts independently establish a game-process delivery path.

TABLE II

REPRESENTATIVE NPC DIALOGUE EVALUATION RESULTS: ALL SEVEN COMPLETED MATCHED QUESTIONS AND THEIR FULL RECORDED RESPONSES. EXPECTED TARGETS ARE LITERAL DIAGNOSTICS; N/R DENOTES NO-RAG/RAG TARGET PRESENCE, NOT SEMANTIC CORRECTNESS.

ID / NPC	Player input and expected target	Observed no-RAG response	Observed RAG response	N/R
D1 / Mira	Where is your forge? Target: northern well	My forge is right here in the village center. State your business or move along.	My forge stands beside the northern well. I repair iron tools there.	no / yes
D2 / Mira	What materials and tools do you repair? Target: iron	I repair standard metal tools and weapons. I don't service anything else.	My forge stands by the northern well where I repair iron tools. I do not sell healing potions.	no / yes
D3 / Iona	Where is your clinic? Target: eastern gate	My clinic is located within the main village dwelling, near the central well. You are welcome to seek assistance there if you are in need of medical care.	My clinic is located behind the eastern gate. I treat burns with silverleaf salve there.	no / yes
D4 / Iona	What salve do you use for burns? Target: silverleaf	I use a soothing calendula and honey mixture for burns. I am not certain if other remedies are kept in the village stores.	I treat burns using silverleaf salve at my clinic. Let me know if you need any assistance.	no / yes
D5 / Elin	What is your inn called? Target: blue lantern	Welcome, traveler. This place is simply known as the village inn.	My inn is the Blue Lantern, located beside the western orchard. We have upstairs rooms to rent if you need a place to stay.	no / yes
D6 / Elin	What soup does your inn serve? Target: barley	Welcome to our inn, stranger. We currently have warm potato soup simmering for our guests.	Welcome to the Blue Lantern. We serve hot barley soup to travellers staying upstairs.	no / yes
D7 / Daren	What documents do you inspect for wagons? Target: permit	I inspect the standard transit permit and cargo manifest for all wagons crossing this bridge. Please present them for review.	My assigned duty at the bridge checkpoint is to inspect wagon permits. I cannot authorize private vault access.	yes / yes

TABLE III

SEVEN MATCHED PAIRS: LITERAL ANSWER-TARGET MENTIONS AND RESPONSE-READY TIME (MS). MENTIONS ARE NOT SEMANTIC-FAITHFULNESS SCORES.

Mode	n	Target	Mean	Median	SD
No RAG	7	1/7	1187.9	1201.4	373.1
RAG	7	7/7	1263.7	1104.9	354.4

Deterministic greetings avoid inference, and source revision is explicit. Warm retrieval was short relative to model response-ready times, but separately collected samples do not establish an end-to-end breakdown or large-corpus scalability alongside a game. Single samples, author-written facts and labels, small corpus size, bundled evidence instructions and quota-dependent stopping limit causal and external validity. Four-persona coverage is not a consistency measurement. Boundary, relationship and continuity cases lack completed model evaluation; no semantic-containment percentage is justified. Network and background-game contention were uncontrolled. Native authored tests complement the controlled study without replacing generated gameplay, voice, perceptual or compatibility evaluation.

TABLE IV

RECORDED NATIVE SKYRIM INTERACTIONS (AUTHORED ROUTE, ZERO MODEL CALLS). CONTEXT OBSERVATIONS AND DELIVERY RECEIPTS HAVE DIFFERENT EVIDENTIAL SCOPE.

NPC	Input	Connector evidence	Delivery result
Jenassa	Hello	Actor and offered-dialogue context	Started; completed
Whiterun Guard	HELLO	Correlated actor/session and text input	Started; completed
Unspecified in receipt file	Greeting trial	Same delivery ID for both receipts	Started; interrupted

VIII. CONCLUSION AND FUTURE WORK

A. Conclusion

Anima addresses the tension between predictable authored NPC dialogue and flexible but potentially unsupported model responses. Authored branches provide control but require content and branching work; unrestricted generation can invent locations, procedures or character knowledge. The implemented architecture combines persona/context appraisal, deterministic engagement, source-bound RAG, provider abstraction and

TABLE V
LOCAL WARM RETRIEVAL LATENCY: 30 OBSERVATIONS PER ROW; MS. VECTOR QUERY DENOTES THE CHROMA QUERY CALL.

Stage	n	Mean	Median	SD
Query embedding	30	16.48	16.59	1.72
Vector query	30	2.17	2.15	0.43
Total retrieval	30	19.96	19.83	1.89

bounded acceptance. Preview and Skyrim interfaces share this executable orchestration rather than separate dialogue engines.

Controlled retrieval ranked the declared relevant source first for all 24 queries over 12 authored documents (P@1, R@3 and MRR: 1.00). Seven matched real-model questions yielded literal answer targets in seven RAG responses and one no-RAG response. The complete outputs expose both evidence-backed details and unsupported alternatives; target presence alone cannot establish correctness. Warm retrieval averaged 19.96 ms across 30 observations. Editing the forge location and synchronizing the index made its replacement retrievable without an application training step, although generated-answer propagation remains unmeasured.

Native Skyrim tests acquired Jenassa's context and correlated authored greeting delivery; a Whiterun Guard trial confirmed uppercase greeting routing with started/completed receipts. A separate interrupted receipt pair exercised lifecycle reporting. Together, these results demonstrate an implemented architecture for externally maintained NPC knowledge and game delivery. Independent persona quality, semantic containment, provider performance equivalence and visible/audible generated gameplay remain unresolved. The practical contribution is a working composition whose measured access and delivery mechanisms can be evaluated separately from language quality.

B. Future Work

Dialogue coverage and evaluation: Extend the eight-question model sample to diverse inputs, repeated persona trials and multiple sessions. Independent blinded raters should assess role consistency, usefulness and claim support with agreement reporting. Relationship and follow-up conditions are necessary because the current single-turn pairs do not test recall or continuity.

Persona and knowledge boundaries: Execute the unreached missing-information, unsupported-premise and private-code cases, then add adversarial role-changing and prompt-injection requests. Measure abstention and claim-level evidence attribution separately from literal answer targets and schema validity. This tests whether authorized evidence is used faithfully rather than merely present in the prompt.

Provider comparison: Run matched repeated conditions on multiple actual backends through the common contract. Compare accepted-response quality, repair/fallback frequency, latency, usage and cost under declared capability differences. Local inference must be benchmarked alongside the game, including resource-constrained deployment, before claiming practical interchangeability.

Retrieval and revision: Evaluate larger game-knowledge collections with independent labels, distractors and multiple relevant passages. Compare hybrid retrieval and reranking against measured failure cases. Measure authorization-filter scaling, incremental indexing and concurrent revisions, then test generated answers after knowledge changes to close the present retrieval-to-answer evidence gap.

Native integration: Expand NPCs and gameplay situations, obtain unobstructed original screenshots, and demonstrate generated dialogue with correlated receipts. Measure audible voice quality, acquisition-to-display/audio latency, interruption behavior, frame times and mod/runtime compatibility. Current authored greetings cannot answer these gameplay-quality questions.

Production operation: Measure context-packing tradeoffs and inference latency under concurrent activity. Extend bounded acknowledged history toward persistent session memory while preserving save/branch isolation, and evaluate durable asynchronous delivery and recovery from ambiguous acknowledgements. Changes should respond to measured failures and resource budgets rather than add an untested autonomous subsystem.

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